

The reduced-order Schmidt–Kalman filter

1. The system and measurement equations are given in Equation (10.56).
2. At each time step execute the following equations to obtain $\hat{x}_k$, the estimate of the desired part of the state:

$$\begin{aligned}
 \tilde{K}_k &= (\tilde{P}_k^- H_1^T + \Sigma_k^- H_2^T) \alpha_k^{-1} \\
 \alpha_k &= H_1 \tilde{P}_k^- H_1^T + H_1 \Sigma_k^- H_2^T + H_2 (\Sigma_k^-)^T H_1^T + H_2 \tilde{P}_k^- H_2^T + R \\
 \hat{x}_k^+ &= \hat{x}_k^- + \tilde{K}_k (y_k - H_1 \hat{x}_k^-) \\
 \tilde{P}_k^+ &= (I - \tilde{K}_k H_1) \tilde{P}_k^- - \tilde{K}_k H_2 (\Sigma_k^-)^T \\
 \Sigma_k^+ &= (I - \tilde{K}_k H_1) \Sigma_k^- - \tilde{K}_k H_2 \tilde{P}_k^- \\
 \tilde{P}_k^+ &= \tilde{P}_k^- \\
 \hat{x}_{k+1}^- &= F_1 \hat{x}_k^+ \\
 \tilde{P}_{k+1}^- &= F_1 \tilde{P}_k^+ F_1^T + Q_1 \\
 \Sigma_{k+1}^- &= F_1 \Sigma_k^+ F_2^T \\
 \tilde{P}_{k+1}^- &= F_1 \tilde{P}_k^+ F_2^T + Q_2
 \end{aligned} \tag{10.67}$$

■ EXAMPLE 10.3

Consider the following system:

$$\begin{aligned}
 x_{k+1} &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} x_k + w_k \\
 y_k &= \begin{bmatrix} 1 & 1 \end{bmatrix} x_k + v_k \\
 w_k &\sim (0, Q) \quad Q = \text{diag}(1, 0) \\
 v_k &\sim (0, R) \quad R = 1
 \end{aligned} \tag{10.68}$$

Figure 10.4 shows a typical example of the estimation error of the first element of the state vector for the full-order filter and the reduced-order filter. It is seen that the performances of the two estimators are virtually identical. In other words, we can save a lot of computational effort with only a marginal degradation of estimation performance by using the reduced-order filter.

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10.4 ROBUST KALMAN FILTERING

The Kalman filter works well, but it assumes that the system model and noise statistics are known. If any of these assumptions are violated then the filter estimates can degrade. This was noted early in the history of Kalman filtering [Soo65, Hef66, Nis66].

Daniel Pena and Irwin Guttman give an overview of several methods of robustifying the Kalman filter [Spa88, Chapter 9]. For example, although the Kalman filter

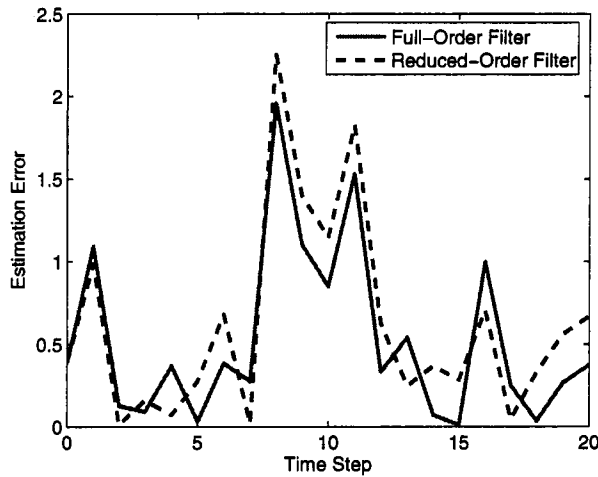


Figure 10.4 Results for Example 10.3. Typical error magnitudes for the estimate of the first state for the full-order filter and the reduced-order filter. The reduced-order filter has only slightly larger estimation errors.

is the optimal linear filter, it is not the optimal filter in general for non-Gaussian noise. Noise in nature is often approximately Gaussian but with heavier tails, and the Kalman filter can be modified to accommodate these types of density functions [Mas75, Mas77, Tsa83]. Sometimes, measurements do not contain any useful information but consist entirely of noise (probabilistically), and the Kalman filter can be modified to deal with this possibility also [Nah69, Sin73, Ath77, Bar78].

The problem of Kalman filtering with uncertainties in the system matrix F_k , the measurement matrix H_k , and the noise covariances Q_k and R_k , has been considered by several authors [Xie94, Zha95, Hsi96, The96, Xie04]. This can be called adaptive filtering or robust filtering. Comparisons of adaptive filtering methods for navigation are presented in [Hid03]. Continuous-time adaptive filtering is discussed in [Bar05, Mar05]. Methods for identifying the noise covariances Q and R are presented in [Meh70, Meh72, Als74, Mye76]. Additional material on robust Kalman filtering can be found in [Che93].

In this section we present a conceptually straightforward way of making the Kalman filter more robust to uncertainties in Q and R [Kos04]. Suppose we have the linear time-invariant system

$$\begin{aligned}
 x_{k+1} &= Fx_k + w_k \\
 y_k &= Hx_k + v_k \\
 w_k &\sim (0, Q) \\
 v_k &\sim (0, R)
 \end{aligned} \tag{10.69}$$

Now suppose that a general steady-state gain K (not necessarily the Kalman gain) is used in a predictor/corrector type of state estimator. The state estimate update equations are then given as follows:

$$\begin{aligned}
\hat{x}_{k+1}^- &= F\hat{x}_k^+ \\
\hat{x}_{k+1}^+ &= \hat{x}_{k+1}^- + K(y_{k+1} - H\hat{x}_{k+1}^-) \\
&= F\hat{x}_k^+ + K(Hx_{k+1} + v_{k+1} - HF\hat{x}_k^+) \\
&= KHx_{k+1} + (I - KH)F\hat{x}_k^+ + Kv_{k+1} \\
&= (KHFx_k + KHw_k) + (I - KH)F\hat{x}_k^+ + Kv_{k+1} \quad (10.70)
\end{aligned}$$

The error in the *a posteriori* state estimate can be written as

$$\begin{aligned}
e_{k+1} &= x_{k+1} - \hat{x}_{k+1}^+ \\
&= (Fx_k + w_k) - [(KHFx_k + KHw_k) + (I - KH)F\hat{x}_k^+ + Kv_{k+1}] \\
&= (I - KH)Fx_k + (I - KH)w_k - (I - KH)F\hat{x}_k^+ - Kv_{k+1} \\
&= (I - KH)Fe_k + (I - KH)w_k - Kv_{k+1} \quad (10.71)
\end{aligned}$$

So the covariance of the estimation error can be written as

$$\begin{aligned}
P_{k+1} &= E(e_{k+1}e_{k+1}^T) \\
&= (I - KH)FP_kF^T(I - KH)^T + (I - KH)Q(I - KH)^T + \\
&\quad KRK^T \quad (10.72)
\end{aligned}$$

The steady-state covariance P satisfies the following Riccati equation:

$$P = (I - KH)FPF^T(I - KH)^T + (I - KH)Q(I - KH)^T + KRK^T \quad (10.73)$$

Note that we derived this without making any assumption on the optimality of the filter gain K . That is, this equation holds regardless of what filter gain K we use. Now we can consider what happens when there is no measurement noise, and what happens when there is no process noise. Define P_1 as the steady-state estimation-error covariance when $R = 0$, and P_2 as the steady-state estimation-error covariance when $Q = 0$. The above equation for P shows that

$$\begin{aligned}
P_1 &= (I - KH)FP_1F^T(I - KH)^T + (I - KH)Q(I - KH)^T \\
P_2 &= (I - KH)FP_2F^T(I - KH)^T + KRK^T \quad (10.74)
\end{aligned}$$

Adding these two covariances together results in

$$\begin{aligned}
P_1 + P_2 &= (I - KH)FP_1F^T(I - KH)^T + (I - KH)Q(I - KH)^T + \\
&\quad (I - KH)FP_2F^T(I - KH)^T + KRK^T \\
&= (I - KH)F(P_1 + P_2)F^T(I - KH)^T + \\
&\quad (I - KH)Q(I - KH)^T + KRK^T \quad (10.75)
\end{aligned}$$

Comparing this equation with Equation (10.73) shows that P and the sum $(P_1 + P_2)$ both satisfy the same Riccati equation. This shows that

$$P = P_1 + P_2 \quad (10.76)$$

This shows an interesting linearity property of a general predictor/corrector type of state estimator. The estimation covariance is equal to the sum of the covariance

due to process noise only and the covariance due to measurement noise only. Recall from Chapter 5 that the Kalman filter was designed to minimize the trace of P . So the Kalman filter minimizes the trace of $(P_1 + P_2)$.

Now suppose that the true process noise and measurement noise covariances are different from those assumed by the Kalman filter. The filter is designed under the assumption that the noise covariances are Q and R , but the true noise covariances are $\tilde{Q}$ and $\tilde{R}$:

$$\begin{aligned}\tilde{Q} &= (1 + \alpha)Q \\ \tilde{R} &= (1 + \beta)R\end{aligned}\tag{10.77}$$

where α and β are unknown scalars. These differences between the assumed and true covariances will result in a change in the estimation-error covariance of the filter. The true estimation-error covariance $\tilde{P}$ will be equal to the assumed covariance P plus some difference ΔP . This can be written as

$$\begin{aligned}\tilde{P} &= (I - KH)F\tilde{P}F^T(I - KH)^T + (I - KH)\tilde{Q}(I - KH)^T + K\tilde{R}K^T \\ P + \Delta P &= (I - KH)F(P + \Delta P)F^T(I - KH)^T + \\ &\quad (1 + \alpha)(I - KH)Q(I - KH)^T + (1 + \beta)KRK^T\end{aligned}\tag{10.78}$$

Comparing this equation with Equation (10.73) shows that

$$\Delta P = (I - KH)F\Delta P F^T(I - KH)^T + \alpha(I - KH)Q(I - KH)^T + \beta KRK^T\tag{10.79}$$

Now we repeat this same line of reasoning for the computation of the true estimation-error covariance when the process noise is zero ($\tilde{P}_1 = P_1 + \Delta P_1$) and the true estimation-error covariance when the measurement noise is zero ($\tilde{P}_2 = P_2 + \Delta P_2$). Equation (10.74) shows that

$$\begin{aligned}\tilde{P}_1 &= (I - KH)F\tilde{P}_1F^T(I - KH)^T + (I - KH)\tilde{Q}(I - KH)^T \\ P_1 + \Delta P_1 &= (I - KH)F(P_1 + \Delta P_1)F^T(I - KH)^T + \\ &\quad (1 + \alpha)(I - KH)Q(I - KH)^T \\ \tilde{P}_2 &= (I - KH)F\tilde{P}_2F^T(I - KH)^T + K\tilde{R}K^T \\ P_2 + \Delta P_2 &= (I - KH)F(P_2 + \Delta P_2)F^T(I - KH)^T + (1 + \beta)KRK^T\end{aligned}\tag{10.80}$$

Comparing these equations with Equation (10.74) shows that

$$\begin{aligned}\Delta P_1 &= (I - KH)F\Delta P_1F^T(I - KH)^T + \alpha(I - KH)Q(I - KH)^T \\ \Delta P_2 &= (I - KH)F\Delta P_2F^T(I - KH)^T + \beta KRK^T\end{aligned}\tag{10.81}$$

Adding these two equations and comparing with Equation (10.79) shows that

$$\Delta P = \Delta P_1 + \Delta P_2\tag{10.82}$$

Comparing Equations (10.74) and (10.81) shows that

$$\begin{aligned}\Delta P_1 &= \alpha P_1 \\ \Delta P_2 &= \beta P_2\end{aligned}\tag{10.83}$$

Combining Equations (10.82) and (10.83) shows that

$$\Delta P = \alpha P_1 + \beta P_2\tag{10.84}$$

Now suppose that α and β are independent zero-mean random variables with variances σ_1^2 and σ_2^2 , respectively. The previous equation shows that

$$\begin{aligned} E[\text{Tr}(\Delta P)] &= E(\alpha)\text{Tr}(P_1) + E(\beta)\text{Tr}(P_2) \\ &= 0 \\ E\{[\text{Tr}(\Delta P)]^2\} &= E\{[\alpha\text{Tr}(X_1) + \beta\text{Tr}(X_2)]^2\} \\ &= \sigma_1^2\text{Tr}^2(P_1) + \sigma_2^2\text{Tr}^2(P_2) \end{aligned} \quad (10.85)$$

This gives the variance of the change in the estimation-error covariance due to changes in the process and measurement-noise covariances. A robust filter should try to minimize this variance. In other words, a robust filter should have an estimation-error covariance that is insensitive to changes in the process and measurement-noise covariances. So the performance index of a robust filter can be written as follows:

$$\begin{aligned} J &= \rho\text{Tr}(P) + (1 - \rho)E\{[\text{Tr}(\Delta P)]^2\} \\ &= \rho[\text{Tr}(P_1) + \text{Tr}(P_2)] + (1 - \rho)[\sigma_1^2\text{Tr}^2(P_1) + \sigma_2^2\text{Tr}^2(P_2)] \end{aligned} \quad (10.86)$$

where ρ is the relative importance given to filter performance under nominal conditions (i.e., when Q and R are as expected), and $(1 - \rho)$ is the relative importance given to robustness. In other words, $(1 - \rho)$ is the relative weight given to minimizing the variation of the estimation-error covariance due to changes in Q and R . If $\rho = 1$ then we have the standard Kalman filter. If $\rho = 0$ then we will minimize changes in the estimation-error covariance, but the nominal estimation-error covariance may be poor. So ρ should be chosen to balance nominal performance and robustness. Unfortunately, the performance index J cannot be minimized analytically, so numerical methods must be used. P_1 and P_2 are functions of the gain K and can be computed using a DARE function in control system software. The partial derivative of J with respect to K must be computed numerically, and then the value of K can be changed using a gradient-descent method to decrease J .

■ EXAMPLE 10.4

Suppose we have a discretized second-order Newtonian system that is driven by an acceleration input. $x(1)$ represents position, $x(2)$ represents velocity, u_k represents the known acceleration input, and w_k represents the noisy acceleration input. This is the same as the system described in Example 9.1. The system is described as follows:

$$\begin{aligned} x_{k+1} &= \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} T^2/2 \\ T \end{bmatrix} u_k + w_k \\ y_k &= \begin{bmatrix} 1 & 0 \end{bmatrix} x_k + v_k \\ w_k &\sim (0, Q) \\ v_k &\sim (0, R) \\ Q &= q^2 \begin{bmatrix} T^4/4 & T^3/2 \\ T^3/2 & T/2 \end{bmatrix} \end{aligned} \quad (10.87)$$

The sample time $T = 0.1$. The variance q^2 of the acceleration noise is equal to 0.2^2 , and the variance R of the measurement noise is equal to 10^2 . Now